

4(a)	<i>allow ideas from a diagram</i> 1 blood arriving, is deoxygenated / has low(er) partial pressure of oxygen A low, concentrations / levels, of oxygen or oxygen-poor or blood arriving has, high(er) partial pressure / AW, of carbon dioxide ; A carbon dioxide-rich 2 <i>idea that</i> newly oxygenated blood is, constantly / continually, removed ; A oxygenated blood passed to pulmonary vein A oxygen that diffuses, from alveoli / into blood, is transported away (by blood flow)	2
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Question	Answer	Marks
4(b)	any two from: allow alveoli to, expand / stretch, on, inhalation / inspiration / breathing in / AW ; prevent alveoli from, overstretching / rupturing / bursting (on, inhalation / inspiration / breathing in) ; I collapse <u>recoil</u> / <u>recoiling</u>, to help, exhalation / expiration / breathing out / force air out ;	2
4(c)	<i>if stated as facilitated diffusion allow max 1 for mp1 or mp2 or mp5</i> any three from: 1 membrane / carrier / transport, protein (move phospholipids across into lamellar body) ; R channel protein / cotransporter (protein) 2 has binding site(s) (specific for surfactant phospholipids) ; R ATP binding site for phospholipids / binds ATP R has receptors to bind to phospholipids 3 ATP, used / hydrolysed / needed, for active transport / to provide energy (for transport) ; A ATP binding site for active transport A ref. to ATPase <u>and</u>, active transport / for energy release I has / presence of, ATP energy 4 (surfactant) phospholipids move, against a concentration gradient / from a low(er) to a high(er) concentration ; 5 able to carry out conformational change / AW ;	3
4(d)(i)	X: transcription ; Y: translation ;	2

Question	Answer	Marks
4(d)(ii)	any three from: allow ecf if introns and exons consistently the wrong way round - introns, are non-coding / do not code for amino acids / are not involved in polypeptide synthesis ; R introns are triplets / codons, that are non-coding I exons are coding - introns are removed, after transcription / from primary transcript / before mRNA is formed ; R introns removed from, <u>m</u>RNA / primary <u>m</u>RNA (only) exons (join to) form mRNA ; A mRNA, contains / AW, exons A only exons are translated <i>if mp 2 <u>and</u> 3 not awarded, allow one mark for:</i> gene splicing / RNA splicing / mRNA does not contain introns or splicing <i>if RNA noted within response – this includes the A for mp4 (UAA etc because these are RNA codons) ;</i> R <u>DNA</u> splicing - some (DNA) triplet(s) / (mRNA) codon(s) are STOP codons ; A UAA / UAG / UGA, is a STOP codon or STOP, codons / triplets, do not code for an amino acid / terminate translation A some, triplets / codons, terminate translation so do not code for amino acids I ref. to premature STOP codons - ref. to met / first amino acid / amino acid coded for by START codon, (may be) removed after translation ; - AVP ; e.g. ref. to (non-coding) regulatory sequences	3